

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458110.7164 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0904 +/- 0.0097
Transit depth (Rp/Rs)^2	0.82 +/- 0.18 [%]
Orbital Inclination (inc)	85.22 +/- 2.2
Airmass coefficient 1 (a1)	1.1598 +/- 0.0083
Airmass coefficient 2 (a2)	-0.142 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	None
Optimal Aperture	4.69
Optimal Annulus	7.82
Transit Duration (day)	0.1145 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0904 ± 0.0097 vs 0.1066 (deviation 1.67σ)
Duration fit vs published	0.1145 d vs 0.1196 d (deviation 1.11σ)
Mid-time O–C	-13.5 min
Depth SNR	9.3
Residual scatter	0.72 %

Light curve

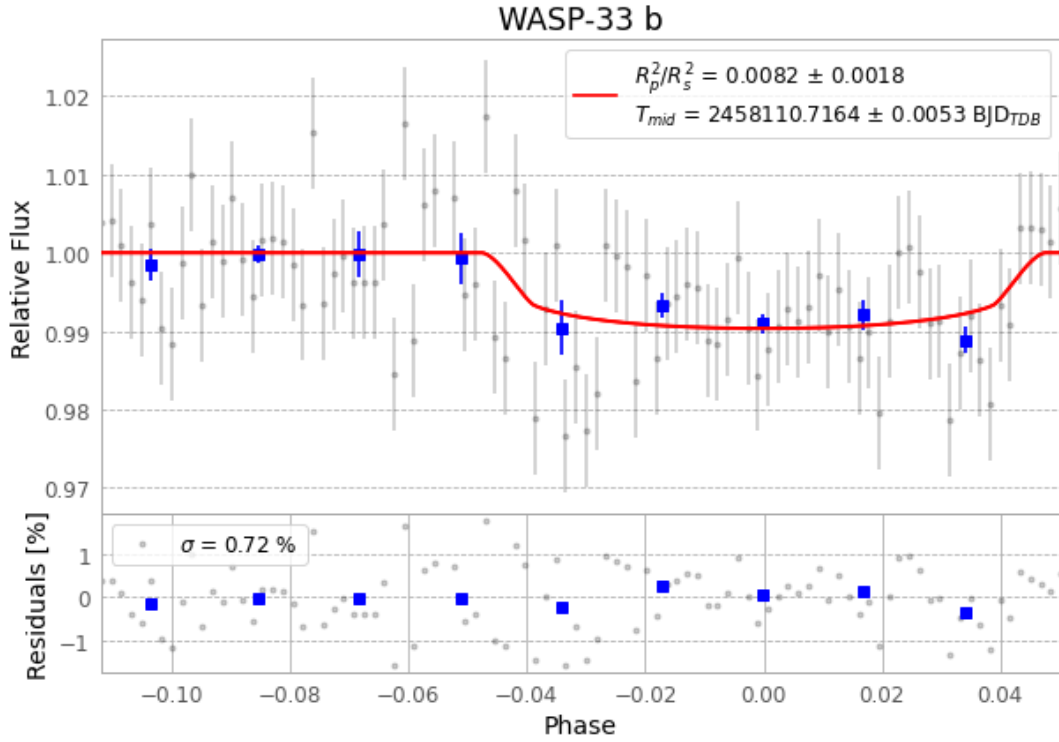


Figure 1 — FinalLightCurve_WASP-33 b_2017-12-22.png (SHA-256 f4be41c5b12eeeb6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd2f01d103b2db02...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a5addef

Data provenance

102 FITS frames (67 MB), WASP-33171223013012.FITS → WASP-33171223063312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-33-171223.log (74873d6dde3b...), FinalLightCurve_WASP-33 b_2017-12-22.png (f4be41c5b12e...), FinalLightCurve_WASP-33 b_2017-12-22.csv (290596a8b704...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 19:07Z.